



Pharm*Achieve*

SAFETY AND DOCUMENTATION

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

LEGEND

ADRs Adverse Drug Reactions

ADE Adverse Drug Event

ADUs Automated Dispensing Units

CARN Canadian Adverse Reaction Newsletter

CADRMP Canadian Adverse Drug Reaction Monitoring Program

CFHI Canadian Foundation for Healthcare Improvement

CIHI Canadian Institute for Health Information

CMIRPS Canadian Medication Incident Reporting and Prevention System

cMAR Computerized Medication Administration Record

CPOE Computerized Provider Order Entry

CPSI Canadian Patient Safety Institute

CQI Continuous Quality Improvement

CVP Canada Vigilance Program

CDSS Clinical Decision Support Systems

EMR Electronic Medical Record

FMEA Failure Mode and Effects Analysis

ISMP Institute of Safe Medication Practices

MAR Medication Administration Record

NCC MERP National Coordinating Council for Medication Error Reporting and Prevention

NSIR National System for Incident Reporting

RCA Root Cause Analysis

MEDICATION PROCESS

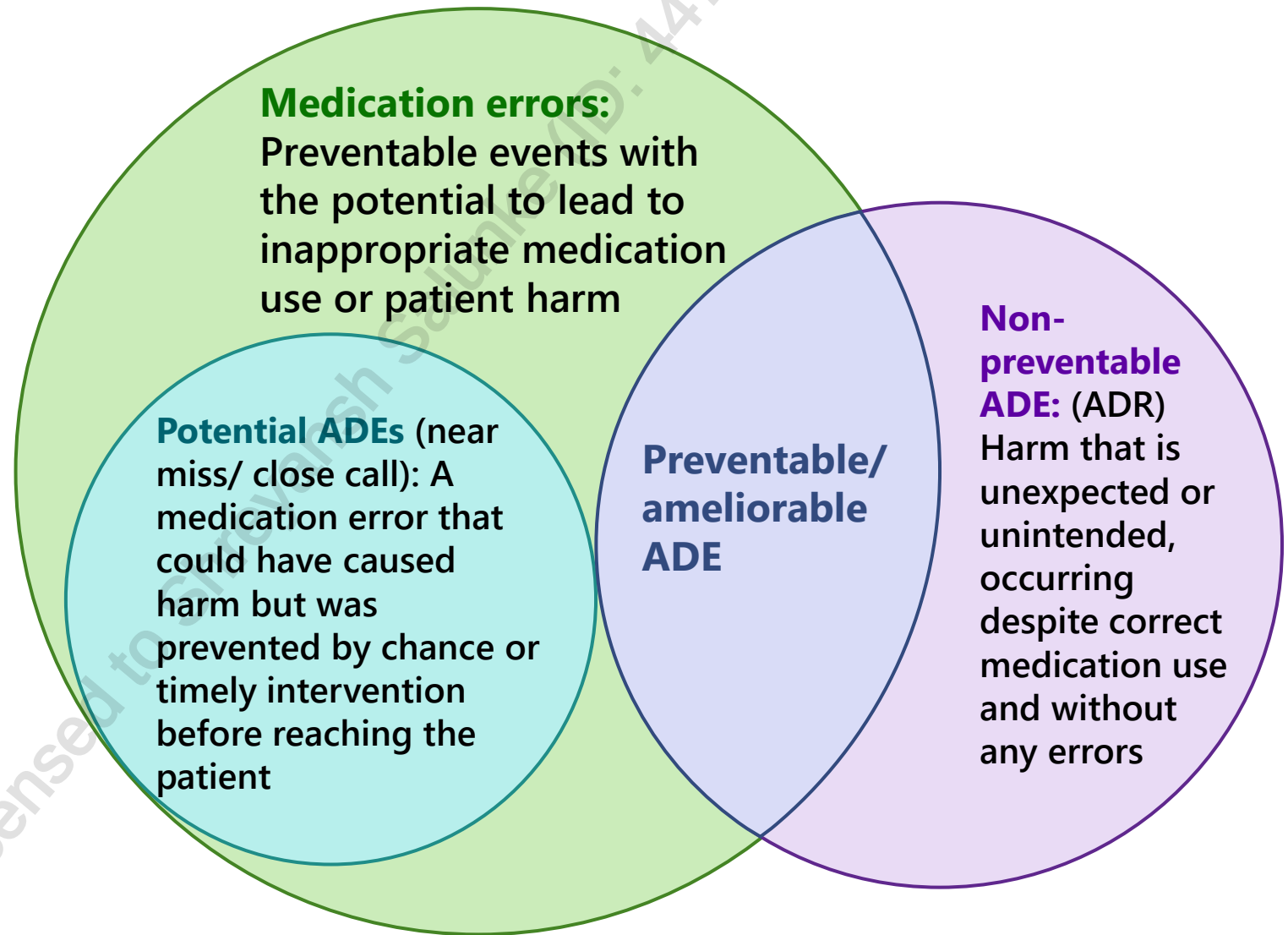
Policies and procedures ensuring safe and effective medication use; consists of 5 stages (*below*)



- Each stage is prone to errors but also has opportunities for preventative measures e.g. independent double checks
- Use of **automation** can reduce errors at most stages
- Medication incident (according to ISMP): Generally, a **preventable mistake** that could cause or contribute to inappropriate medication use or patient harm
 - Incidents do not need to reach the patient
 - Medication error is also referred to as a medication incident
 - Common medication incidents can involve prescribing, professional practice, drug products, procedure product labelling, packing, professional practice, drug products, order communication, compounding, dispensing, administration, education, monitoring, nomenclature or the system
 - Example: dispensing the wrong medication to a patient

MEDICATION INCIDENT

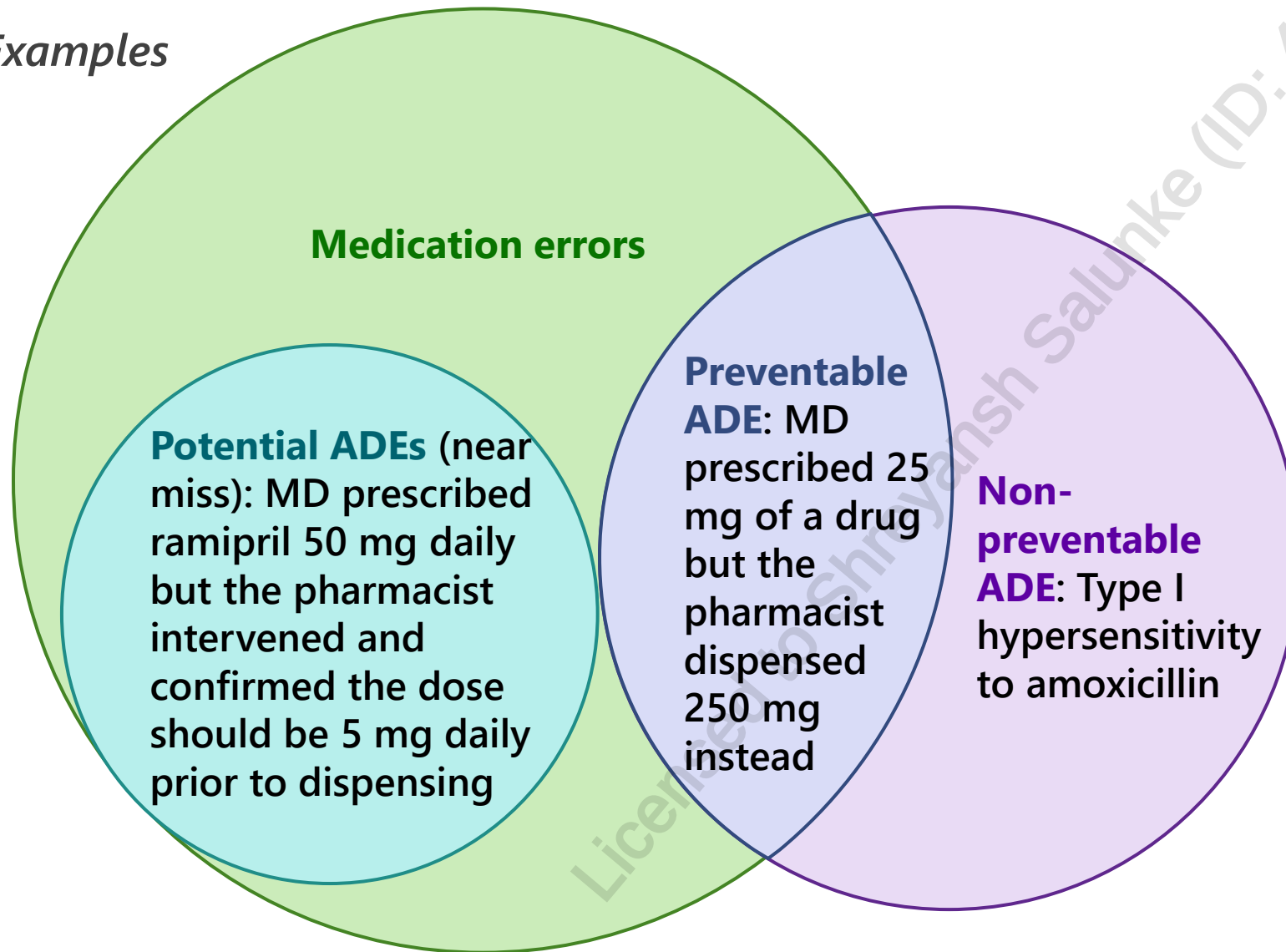
ADE: harm from medication or from not receiving one, including adverse drug reactions and medication incidents



What's the difference between ADRs and side effects?

MEDICATION INCIDENT

Examples



Factors for errors?

- Prescribing: poor handwriting, verbal order
- Dispensing (packaging, labelling, compounding): drug interactions, polypharmacy
- Administration: incorrect scheduling
- Counselling
- Communication: look-alike and sound-alike drugs
- Work environment: Increased workload, insufficient

TYPES OF PATIENT SAFETY INCIDENTS

As Defined by ISMP:

1

Near Miss

- Does not reach the patient

2

Harmful Incident

- Reaches the patient and results in harm

3

No-Harm Event/Incident

- Reaches the patient and does NOT result in harm

CANADA VIGILANCE PROGRAM

- Gathers and evaluates **reports of suspected adverse reactions** to health products sold in Canada:
 - Prescription and non-prescription medications
 - Natural health products
 - Biologically derived products such as vaccines and fractionated blood products, cells, tissues and organs
 - Radiopharmaceuticals
 - Disinfectants and sanitizers with disinfectant claims
- Can be submitted by health professionals and consumers on a **VOLUNTARY** basis
- Reporting can be done online, by phone or by submitting the Side Effect Reporting Form by fax or mail
- Market Authorization Holders (manufacturers and distributors) are **REQUIRED to report** adverse reactions as mandated by the Food and Drugs Regulations, the Natural Health Product Regulations and the Safety of Human Cells, Tissues and Organs for Transplantation Regulations
- Enables Health Canada to ensure the **benefits of the marketed products continue to outweigh the risks**



Canada's post-marketing surveillance program

MEDEFFECT CANADA

Developed by the Marketed Health Products Directorate to:

- **Provide centralized access to relevant and reliable health product safety information**
- Includes access to Health Canada's advisories, warnings and recalls; the Canadian Adverse Reaction Newsletter (CARN); and the Canadian Adverse Drug Reaction Monitoring Program (CADRMP) Online Query and Data Extract
- Make it as simple and efficient as possible for health professionals and consumers to complete and file adverse reaction reports via web, phone, fax or mail
- Learn and better understand the importance of reporting side effects



Government of Canada | MedEffect Canada

CMIRPS

CANADIAN MEDICATION INCIDENT REPORTING AND PREVENTION SYSTEM

A voluntary confidential program collecting, analyzing, and distributing information regarding medication incidents to ultimately make medication use safer; key partners-Health Canada, ISMP Canada, CIHI, CPSI.

Anyone can report medication incidents to CMIRPS to help improve the safety of medication use in Canada. There are multiple channels (each for different class of user) for confidential/secure reporting:

1. **Consumers and patients: ISMP SafeMedicationUse.ca** collects medication incident reports (online/telephone) from consumers and provides resources to help prevent incidents

*** Does NOT collect ADR/side effect (report to different channels i.e. CVP)

2. **Healthcare facilities: CIHI National System for Incident Reporting (NSIR)** is a voluntary reporting system for *institutions* to record medication incidents data. Designed to make it easier to share and to learn from medication incidents at both the local and system levels

3. **Individual practitioners: Individual Practitioner Reporting System** collects and analyzes medication incidents reported (online or by telephone) by individual practitioners in a confidential channel that can serve as an early warning system. Reports are accepted online

4. **Community pharmacies: National Incident Data Repository for Community Pharmacies** captures and analyzes medication incidents submitted by community pharmacies to create a cohesive information-sharing system for better understanding of medication incidents and development of more robust strategies to prevent harm

MEMORIZATION AID- WHERE DO YOU REPORT?

WHAT IS REPORTED?

Medication Incidents

Medication Errors
(are one type of incident)

Adverse Drug Effects

OR

New Health Product
Safety Concerns

WHERE IS IT REPORTED?

Canada
Medication
Incident
Reporting
and
Prevention
System
(CMIRPS)

Run by
ISMP
Canada

Run by
CIHI

SafeMedicationUse.ca

National Incident Data
Repository for Community
Pharmacies

Individual Practitioner
Reporting System

National System for
Incident Reporting

Canada Vigilance
Program

MedEffect Web Site
(operated by Health
Canada)

BY WHOM?



Consumers



Community
Pharmacies



Individual
Practitioners



Institutions



ANYONE
(Health
Professionals,
Consumers)

CMIRPS KEY PARTNERS

ISMP	Independent, national not-for-profit organization committed to advancing medication safety in all healthcare settings through analysis, recommendations and facilitating quality improvement measures; SafeMedicationUse.ca
CIHI-NSIR	Independent, not-for-profit organization providing information and data on Canada's healthcare system (including performance) and health of Canadians (including population health), to accelerate improvements
CPSI	- Not-for-profit organization existing to raise awareness and to facilitate implementation of strategies for safer healthcare; - Provides checklists and tools for patients and providers (e.g. stewardship); - Works with CMIRPS to provide marketing and communication support (e.g. Global Patient Safety Alerts) and to ensure CMIRPS activities are coordinated with other patient safety initiatives in Canada - Publishes Safer Healthcare Now! (website and campaign that provides resources for frontline healthcare providers, delivery systems and others to improve patient safety) - Legally amalgamated with Canadian Foundation for Healthcare Improvement (CFHI) as a new organization, Healthcare Excellence Canada

VANESSA'S LAW

- As of December 16, 2019, under the *Protecting Canadians from Unsafe Drugs Act*, it is **mandatory** for hospitals to **report serious adverse drug reactions and medical device incidents** to Health Canada
- Applies to therapeutic products including prescription and over-the-counter drugs, biologics (biotechnology products, blood products, plasma proteins and vaccines*), radiopharmaceutical drugs, disinfectants and medical devices.
- Intended to improve Health Canada's ability to collect post-market safety information and take appropriate action when a serious health risk is identified

*Excludes vaccines administered under a routine immunization program of a province or territory

VANESSA'S LAW

Serious adverse drug reaction

- a harmful and unintended response to a drug that occurs at any dose leading to:

- A death or life-threatening situation,
- A disability,
- An admission to hospital or lengthened hospital stay,
- A congenital malformation, or
- A medical intervention to avoid any of the above

Medical device incident

- incident related to a failure of a medical device (ex. decline in its effectiveness, any inadequacy in its labelling or directions for use) that led or could have potentially led to death or serious deterioration in the health of the patient, user or other person such as:

- A life-threatening disease, disorder or abnormal physical state
- A permanent impairment to body function or structure,
- A disorder or abnormal physical state or permanent impairment or damage
- A medical intervention to avoid any of the above

CASE EXAMPLE 1

Is the hospital required to report?

A patient on medications, including warfarin, presents to the emergency department with a life-threatening gastrointestinal bleed. The patient had to be hospitalized to ensure they were stabilized.

CASE EXAMPLE 2

Is the hospital required to report?

A patient diagnosed with Hodgkin's lymphoma was being treated with doxorubicin, bleomycin, vincristine, and dacarbazine. After cycle 3, the patient was admitted to the hospital. They complained of a dry cough and shortness of breath on exertion. Bleomycin-induced pulmonary fibrosis was suspected.

CASE EXAMPLE 3

Is the hospital required to report?

A patient was recently started on the oral anticoagulant warfarin and is having their international normalized ratio (INR) monitored at a an out-patient anticoagulation clinic at a hospital. The patient reported a nosebleed that occurred between clinic appointments. Based on the patient's INR level, the patient's warfarin dose was adjusted. The patient will continue to have their INR monitored at the hospital.

CASE EXAMPLE 4

Is the hospital required to report?

A patient being treated with doxorubicin and cyclophosphamide has developed neutropenia. Upon assessing the severity of the neutropenia, it was determined that the patient may continue with chemotherapy at a reduced dose with growth factor support.

CASE EXAMPLE 5

Is the hospital required to report?

A patient reported feeling dizzy and sweating after taking a dose of insulin. The patient needed glucose tablets to recover. It was identified that a short-acting insulin was provided to the patient instead of the long-acting insulin they normally take.

CASE EXAMPLE 6

Is the hospital required to report?

Multiple patients undergoing endometrial ablation of the uterus suffered burns to adjacent organs. The burns were a result of thin uterine walls and an unintended side effect of the ablation. It was discovered that the manufacturer failed to change the ablation device label to warn users of this side effect, which may occur even when the device is working within its specification.

CASE EXAMPLE 7

Is the hospital required to report?

A health care professional reported that the sewing cuff was defective during a heart valve implant. The defective valve was discarded and a new valve was implanted. Pumping time during the surgery was extended. The defect had the potential to cause serious harm.

CASE EXAMPLE 8

Is the hospital required to report?

A manufacturer released a batch of out-of-specification blood glucose test strips. A patient uses the test strips according to the manufacturer's instructions. However, the readings given were incorrect, which led to improper insulin dosing, hypoglycemic shock and hospitalization.

CASE EXAMPLE 9

Is the hospital required to report?

A health care provider performed an inflation test before inserting a balloon catheter into a patient, as required by the instructions for use that came with the device. A malfunction on inflation was detected and another balloon was used.

CASE EXAMPLE 10

Is the hospital required to report?

A patient died after dialysis treatment. The patient had end-stage renal disease and died of renal failure.

ERROR-REDUCTION TECHNOLOGIES

CPOE: Structured computer order-entry system that ensures all components of prescribing (e.g. dose, frequency) are entered; reduces errors related to poor handwriting or transcription. Checks drug orders against C/Is and allergies in patient record.

CDSS: IT knowledge systems using patient data to improve clinical decision-making by using patient characteristics with computer algorithms to generate case-specific recommendations e.g. drug therapy recommendations based on blood work

Robots (filling prescriptions): Reduces dispensing errors in pharmacies (e.g. wrong drug)

Bar coding: Ensures the right patient, right drug, right route, right time, right dose, etc.

eMAR (computerized MAR, cMAR): Report of drugs administered to a patient by a health care provider; part of patient's permanent medical record; electronic version reduces discrepancies

ADUs: Computerized point-of-use machines that dispense drugs, allowing clinicians access to patient profiles on hospital wards. Medication access is tracked based on patients' profiles

Smart Pumps: Infusion pumps with dose calculation software that identify and correct pump-programming errors. Particularly with high-hazard drugs, it allows clinicians to pre-program standard concentrations, upper and lower dose limits. These pumps are programmed to warn if the dose is too high

How are barcoding, ADU devices, and cMAR used in hospitals or other settings?

DRUG MONITORING

- Computerized detection and alert of adverse drug reactions, drugs with narrow therapeutic index, and toxic serum levels
 - E.g. medication monitoring software- minimizes adverse drug events occurring during the monitoring stage of the medication use process
 - Examples include ECG for QT prolongation
- *What are some **advantages** and **disadvantages** of automation?*

Technologies to increase patient safety

TRUE OR FALSE

When a medication error occurs, the most important step is to educate the person involved in the error.

REDUCING ERRORS

From an institution's perspective

- Promote incident reporting, and no-blame culture
- Technologies for quality improvement: CPOE, CDSS, EMR, computerized MAR, use of barcodes on patient bracelets and on drug bottles.
- Regular policy and procedure reviews and updates
- Conduct staff training on error prevention and reporting
- Implement strategies preventing error recurrences
- Analyze medication error data using Failure Mode and Effects Analysis (FMEA) or Root Cause Analysis (RCA)

From a pharmacist/technician's perspective

- Follow ISMP safety bulletins and provide appropriate patient education
- Interprofessional collaboration to implement error-prevention strategies
- Use TALLman lettering and dispensing technologies (e.g. robots)
- Report errors and near misses through internal (hospital) and external reporting systems

From a nurse's perspective

- Use unit-dose systems instead of traditional ward stock
- Identify patients using full name, date of birth, and medical record number
- Use cMAR

From an MD's perspective

- Avoid abbreviations, write legibly, avoid verbal orders
- Use pre-printed orders and interprofessional collaboration (e.g. pharmacists, medication reconciliation tools)
- Use Electronic Medical Records (EMRs) to avoid handwriting, CPOE (to check prescriptions)

ISMP'S DO NOT USE LIST

Error-prone abbreviations, symbols, and dose designations that should be avoided in medical settings

Abbreviation	Intended Meaning	Problem	Correction
U	unit	Mistaken for "O" (zero), "4" (four), or cc	Use "unit"
IU	international unit	Mistaken for "IV" (intravenous) or "10" (ten)	Use "unit"
Abbreviations for drug names		Misinterpreted because of similar abbreviations for multiple drugs; e.g., MS, MSO ₄ (morphine sulphate), MgSO ₄ (magnesium sulphate) may be confused for one another	Do not abbreviate drug names
QD QOD	Every day Every other day	QD and QOD have been mistaken for each other, or as 'qid'. The Q has also been misinterpreted as "2" (two)	Use "daily" and "every other day"
OD	Every day	Mistaken for "right eye" (OD = oculus dexter)	Use "daily"
OS, OD, OU	left eye, right eye, both eyes	May be confused with one another	Use "left eye", "right eye" or "both eyes"
D/C	Discharge	Interpreted as "discontinue whatever medications follow" (typically discharge medications)	Use "discharge"
cc	cubic centimetre	Mistaken for "u" (units)	Use "mL" or "millilitre"
µg	microgram	Mistaken for "mg" (milligram) resulting in one thousand-fold overdose	Use "mcg"

ISMP'S DO NOT USE LIST

Error-prone abbreviations, symbols, and dose designations that should be avoided in medical settings

Symbol	Intended Meaning	Potential Problem	Correction
@	at	Mistaken for "2" (two) or "5" (five)	Use "at"
> <	Greater than Less than	Mistaken for "7"(seven) or the letter "L". Confused with each other	Use "greater than"/"more than" or "less than"/"lower than"
Dose Designation	Intended Meaning	Potential Problem	Correction
Trailing zero	$x.0$ mg	Decimal point is overlooked resulting in 10-fold dose error	Never use a zero by itself after a decimal point. Use " x mg"
Lack of leading zero	$.x$ mg	Decimal point is overlooked resulting in 10-fold dose error	Always use a zero before a decimal point. Use " $0.x$ mg"

TALLMAN LETTERING (ISMP)

Method of differentiating unique-letter characteristics of similar drug names by capitalizing differences to avoid confusion.

dimenhy**DR**INATE/diphenhy**DR**AMINE

predni**S**ONE/predniso**L**ONE

HIGH-ALERT MEDICATIONS (ISMP)

Implement safeguard measures e.g. separate storage, limited access, highly visible auxiliary labels, independent double checks

Anesthetics and sedation agents (oral and intravenous) e.g. ketamine, propofol

Antithrombotic agents
e.g. oral/injectable anticoagulants, thrombolytics

Chemotherapeutic/Antineoplastic medications
e.g. cisplatin, cyclophosphamide, methotrexate

Electrolyte and related intravenous solutions
e.g. calcium (all salts) with concentration $\geq 10\%$,
magnesium sulfate with concentration $\geq 20\%$

Insulin products

Neuromuscular blockers/Paralyzing agents
e.g. rocuronium

Opioids

e.g. fentanyl, hydromorphone, hydrocodone

Medications with profound cardiovascular/physiologic effects

e.g. IV adrenergic agonists/antagonists, IV antiarrhythmics

Medications for specific circumstances

e.g. medications administered via epidural or intrathecal route

Specific medications or therapies

e.g. injectable benzodiazepines, methotrexate when used for non-cancer indications

Drugs with high risk of harm or death if inappropriately administered (e.g. wrong dose or patient).



This term does **NOT** mean that more errors/confusions occur with these drugs

3 COMMON DISPENSING MODELS

	TRADITIONAL	AUTOMATED DISPENSING UNITS	UNIT-DOSE
Description	Medications are dispensed in bulk into bins in medication room and are used by nurses as needed	Each medication dispensed in different locked cabinet. Computer unlocks only cabinet with ordered drug	Medications are dispensed into ready to administer packages specific to a patient
For whom does the pharmacy dispense?	Entire unit, floor or ward 	A specific unit or ward 	A specific patient 
Safety	LOW Since medications are not checked by pharmacy against patient profiles of a specific patient	MEDIUM Better than traditional since AD machine only allows cabinet with correct medication to be opened but dose not checked by pharmacy	HIGH Since pharmacy checks each patient's dispensed medication against patient profile
Replenishment cycle from pharmacy	Once per week for patient specific items but q24h for stock items	Every 12 hours	Every 24 hours

SAFETY CULTURE

- The shared beliefs, values, norms, and repeated behaviours prevalent in an organization
- It is the “how we do things, how we think and what we value around here” concepts
- The “code” that runs the company
- Guides how decisions are made, and what is prioritized in the absence of supervision
- A safety culture is the prerequisite for information
 - In a punitive culture, information hides. In a safety culture, information flows
 - When employees feel ‘psychologically safe’, they report

Why culture? Why not just policies and supervision?

- Policies can not predict every single possible scenario, and supervision does not scale
- A safety culture closes the gap between “Work as Imagined” (policies/manuals/protocols in place) and “Work as Done” (what *actually* happens)
- A safety culture is a culture of integrity, **where safety protocols are followed even when no one is watching**

How Do We Build A Safety Culture?

GOAL: An “Informed” Culture: where everyone understands and recognizes the hazards in the operating environment

The 3 Components of a Safety Culture:

I. Just culture

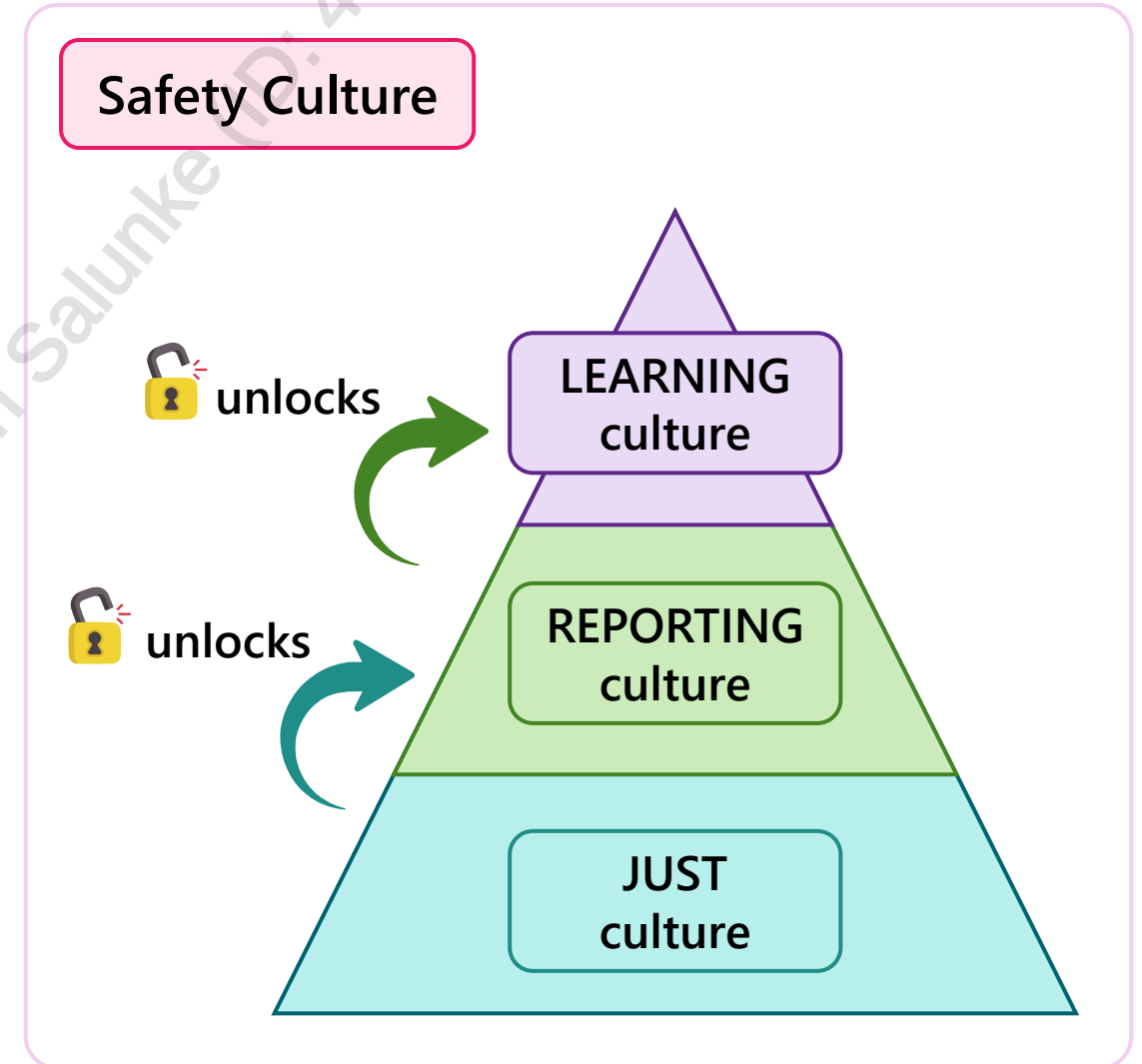
- A culture of trust and accountability: Honest mistakes are supported; reckless risks and deliberate violations are disciplined

II. Reporting culture

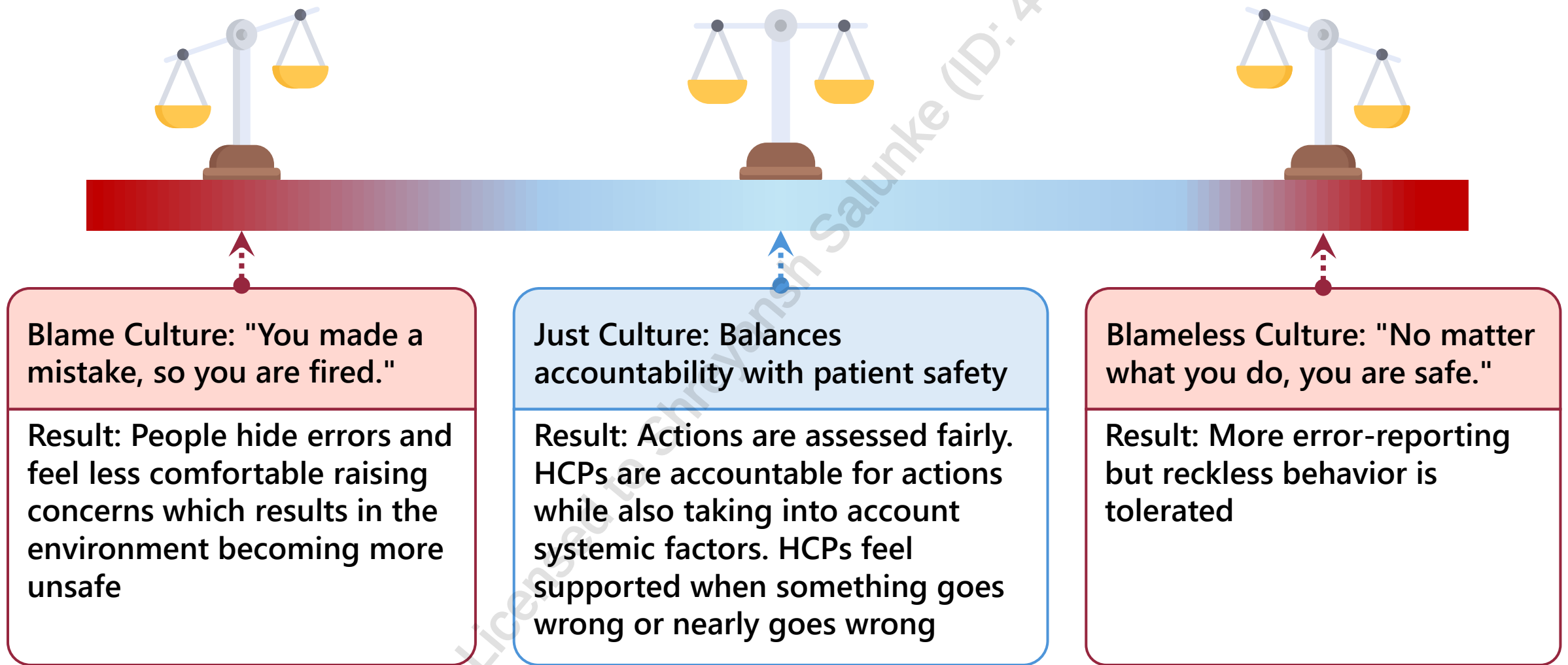
- Willingly reporting practice incidents and near misses without fear of retribution

III. Learning culture

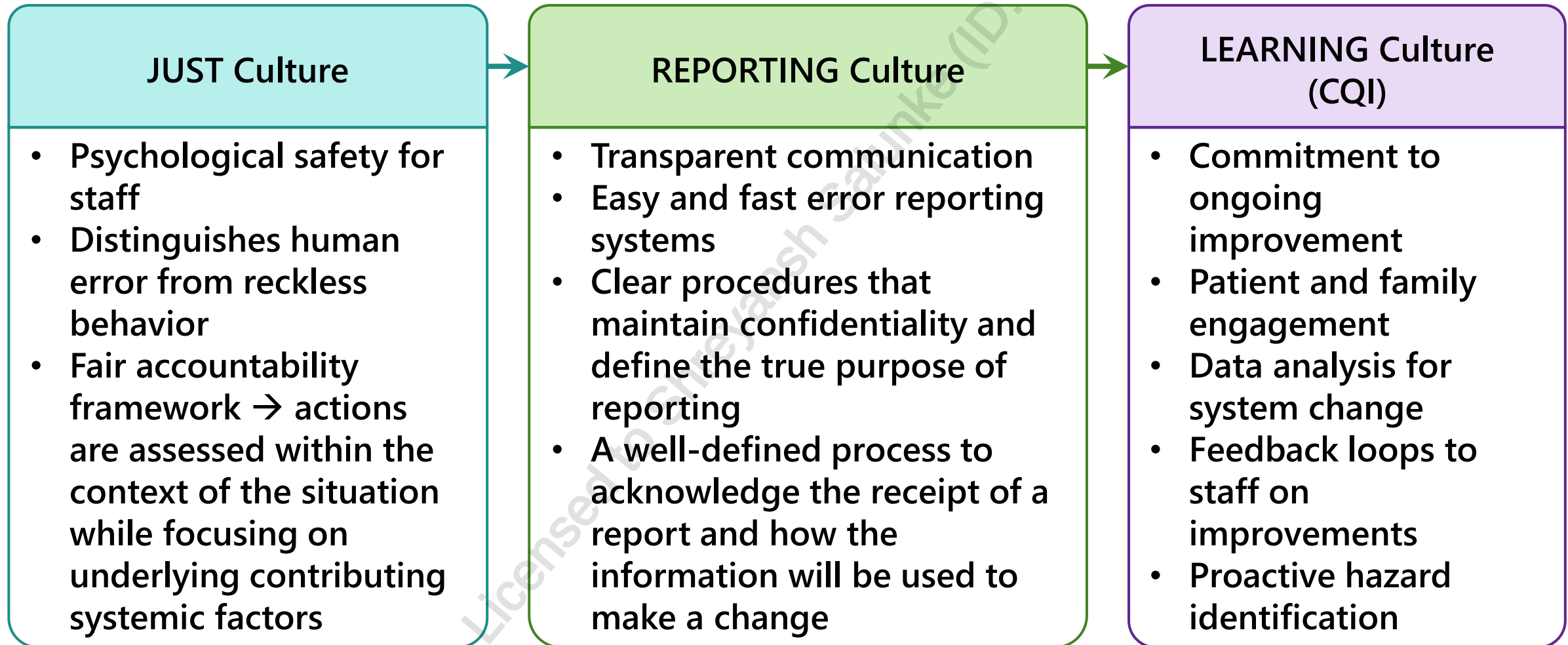
- Continuous learning and improvement
- Proactive state of mind to prevent future harm



What is a Just Culture?



Building a Safety Culture – The Building Blocks



TAKEAWAY

- **Canada Vigilance Program:** collects and assesses reports of suspected adverse reactions to health products marketed in Canada. **Voluntary** for health care professionals and consumers to report. **Mandatory** for manufacturers and distributors to report.
- **MedEffect Canada:** portal to access to relevant and reliable health product safety information
- **CMIRPS:** **voluntary** confidential program collecting, analyzing, and distributing information regarding medication incidents
- **ISMP:** collects medication incident reports and provides resources to help prevent incidents
- **Vanessa's Law:** **mandatory** for hospitals to report serious adverse drug reactions and medical device incidents to Health Canada
- **Document, document, document!**
- Medication incident reporting is NOT designed to be punitive but rather **CONSTRUCTIVE**
 - Focuses on the incident and factors related as opposed to the person
 - Helps to fix **SYSTEMIC** issues to reduce the risk of future errors
 - Essential to assist management in improving patient safety
- **Safety culture:** the shared beliefs, values, norms, and repeated behaviours prevalent in an organization
 - A **JUST** culture unlocks a **REPORTING** culture, which unlocks a **LEARNING** culture

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CHANGE LOG

March 2025

- Content reviewed, no changes made

August 2025

- Content reviewed; no major changes made
- Slides 4,5 and 8: Rephrasing changes made

November 2025

- Content reviewed; no major changes made

January 2026

- Slide 2: Updated NAPRA competencies slide to align with the 2024 competencies
- Legend: Updated CPOE to Computerized Provider Order Entry
- Slides 33-36: Added information on Safety Culture
- Slide 37: Updated takeaway
- Slide 38 Updated references